

QUESTION 1

What is the sum of binary numbers 101101_2 and 110001_2 ?

- A. 1011110_2 B. 1101101_2 C. 1010110_2 D. 1110010_2 E. 1101010_2

QUESTION 2

What is output by the code to the right?

- A. No B. YesNo
C. Yes D. NoYes
E. Nothing

```
int x = 10;
if (x < 100)
    System.out.print("Yes");
else
    System.out.print("No");
```

QUESTION 3

What replaces **<*1>** in the code to the right as the name of the static method that is executed when MyClass is executed by the Java interpreter?

- A. start B. init
C. main D. void
E. first

```
public class MyClass {
    public static void <*1>(String[] args) {
        // code not shown
    }
}
```

QUESTION 4

What does `int[] intArray` look like after the static method call `process(intArray)` when `intArray` begins as the array below?

1	2	3	4	5	6
---	---	---	---	---	---

- A.

1	2	3	4	5	6
---	---	---	---	---	---

B.

3	5	7	9	11	13
---	---	---	---	----	----

C.

3	6	9	12	15	18
---	---	---	----	----	----

D.

4	5	6	7	5	6
---	---	---	---	---	---

E. An exception is thrown

```
public static void process(int[] a) {
    for (int i=0; i < a.length; ++i)
        a[i] = a[i] * 2 + 1;
}
```

QUESTION 5

Which of these types is used to represent floating point numbers?

- A. int B. character C. double D. String E. ArrayList

QUESTION 6

What replaces **<*1>** in the code to the right to indicate that `currentId` is a class variable shared by all instances of `Employee` and hidden from other classes?

- A. `hidden static`
- B. `public static`
- C. `static`
- D. `protected static`
- E. `private static`

```
public class Employee {
    public Employee(String name) {
        this.name = name;
        this.id = currentId++;
    }
    private String name;
    private long id;
    <*1> int currentId = 0;
}
```

QUESTION 7

Assume **<*1>** is filled in correctly. What will be the `id` of the second `Employee` created by a program?

- A. 0
- B. 1
- C. 2
- D. 3
- E. Cannot be determined

QUESTION 8

How many `*`'s are output by the code to the right?

- A. 10
- B. 0
- C. 1
- D. 100
- E. 99

```
int x = 10, y = 100;
int z = y % x;
do System.out.print('*');
while (--z > 0);
```

QUESTION 9

Which of these expressions returns the value `"00"`?

- A. `s.substring(2,3)`
- B. `s.substring(1,3)`
- C. `s.substring(1,2)`
- D. `s.substring(2,2)`
- E. `s.charAt(1) + s.charAt(1)`

```
String s = "2006 UIL Regional";
```

QUESTION 10

Suppose `x`, `y`, and `z` are initialized to 1, 2, and 3, respectively. How many of the comparison operations are evaluated to get the value of `b`?

- A. 2
- B. 3
- C. 4
- D. 5
- E. 6

```
int x, y, z;
// code to initialize x, y, and z not shown
boolean b =
    ((x < y) && (z < y)) ||
    ((y < z) || (z < x)) ||
    ((x >= z) && (y <= z));
```

QUESTION 11

What replaces **<*1>** in the code to the right to call the static method in the `Math` class that computes square roots to get the square root of `i`?

- A. `Math m = new Math().sqrt(i)`
- B. `Math.sqrt(i)`
- C. `Math.root(i, 2)`
- D. `Math.root(i, 1/2)`
- E. `new Math().root(i, 0.5)`

QUESTION 12

Assume **<*1>** is filled in correctly. Assume `i` is positive and `r` is bigger than 2. What is the running time of `root()`? Choose the most restrictive correct answer.

- A. $O(i*r)$
- B. $O(i^r)$
- C. $O(r)$
- D. $O(r^{1/i})$
- E. $O(i^{1/r*r})$

```
public static int findRoot(int i, int r) {
    if (r == 2)
        return (int) <*1>;
    else {
        for (int j=0; ;++j) {
            int n = 1;
            for (int k=0; k<r; ++k) {
                n *= j;
            }
            if (n > i) return j-1;
        }
    }
}
```

QUESTION 13

What replaces **<*1>** in the code to the right to give a random value between 1 and `n`, inclusive?

- A. `r.nextInt(n) + 1`
- B. `r.nextInt(n)`
- C. `r.next(int) % n`
- D. `r.nextInt() % n + 1`
- E. Either A or D

```
public class Dice {
    public Dice(int numSides) {
        n = numSides;
    }
}
```

QUESTION 14

Assume **<*1>** is filled in correctly. Which of these is the most likely to be output by the code below?

```
Dice d1 = new Dice(6),
    d2 = new Dice(10);

System.out.print(d1.roll() +
                d2.roll());
```

- A. 5
- B. 8
- C. 11
- D. B and C are equally likely
- E. A, B, and C are equally likely

```
public int roll() {
    return <*1>;
}

private int n;
private static Random r = new Random();
}
```


QUESTION 15

Which of these replaces **<*1>** in the code to the right to initialize data member `digits` to a byte array with length equal to `length`?

- A. `digits.length() = length`
- B. `digits.length = length`
- C. `digits[length] = new byte[]`
- D. `digits.length = new byte[length]`
- E. `digits = new byte[length]`

QUESTION 16

Which of these replaces **<*2>** in the code to the right to set `i` to 0 if `neg` is false and 1 if `neg` is true?

- A. `1 - neg`
- B. `(neg?1:0)`
- C. `neg`
- D. `(neg?0:1)`
- E. More than one of these

For the remaining questions, assume that **<*1>** and **<*2>** have been filled in correctly.

QUESTION 17

What value is `digits` initialized to if the first constructor is called with parameter 0?

- A. `0`
- B. `{0}`
- C. `null`
- D. `{}`
- E. A run-time error occurs

QUESTION 18

Which of these best describes the way `-38` would be stored in a `MyInteger` object?

- A. In an array of length 2 with `-8` in the first position and `-3` in the second position
- B. In an array of length 2 with `-3` in the first position and `8` in the second position
- C. In an array of length 2 with `8` in the first position and `-3` in the second position
- D. In an array of length 2 with `-3` in the first position and `-8` in the second position
- E. As in answer A if created by the `int` constructor, but as in answer C if created by the `String` constructor

```
public class MyInteger {
    public MyInteger(int i) {
        if (i != 0) {
            int numDigits = 0, j = i;
            do {
                numDigits++; j/=10;
            } while (j != 0);
            digits = new byte[numDigits];
            for (int k=0; k<numDigits; ++k) {
                digits[k] = (byte)(i%10);
                i/=10;
            }
        }
    }

    public MyInteger(String s) {
        if (!s.equals("0")) {
            boolean neg = false;
            if(s.charAt(0) == '-') neg = true;
            int length = s.length();
            if (neg) --length;
            <*1>;
            for (int i = <*2>; i < s.length(); ++i) {
                char ch = s.charAt(i);
                if (ch < '0' || ch > '9')
                    throw
                        new IllegalArgumentException();
                digits[s.length()-i-1] =
                    (byte)((ch-'0')*(neg?-1:1));
            }
        }
    }

    public String toString() {
        if (digits == null) return "0";
        else {
            String s = "";
            if (digits[0]>0)
                for (int i = digits.length-1; i>=0; --i)
                    s += (char)(digits[i]+'0');
            else {
                s += '-';
                for (int i = digits.length-1; i>=0; --i)
                    s += (char)(-digits[i]+'0');
            }
            return s;
        }
    }

    // arithmetic methods not shown

    private byte[] digits;
}
```

QUESTION 19

What is the output of the code to the right on the input below?

134 569 2abc

- A. 1345692Not an integerErrorDone
- B. 134
- C. 134Done
- D. 2Error
- E. No output

```
// nextInt() throws InputMismatchException
// when the next token is not an integer
```

```
Scanner in = new Scanner(System.in);
try {
    int x = in.nextInt();
    System.out.print(x);
}
catch(InputMismatchException e1) {
    System.out.print("Not an integer");
}
catch(Exception e2) {
    System.out.print("Error");
}
finally {
    System.out.print("Done");
}
```

QUESTION 20

What is the output of the code to the right on the input below?

cba2 965 431

- A. Not an integerDone
- B. cbaNot an integerErrorDone
- C. 2
- D. 2Done
- E. Not an integerErrorDone

```
public static <E> int
search(ArrayList<E> list, E item) {
    for (int i=0; i<list.size(); ++i)
        if (<*1>)
            return i;
    return -1;
}
```

QUESTION 21

What expression replaces <*1> in the code to the right to check whether the item at position *i* in *list* is the same as the *item* parameter?

- A. list[i] == item
- B. list.get(i).equals(item)
- C. list[i].equals(item)
- D. ((E)list.get(i)).equals<E>(item)
- E. item.equals(list[i])

QUESTION 22

Assume <*1> is filled in correctly. What is the running time of `search(li, it)` where *li* contains *n* items, none of which is *it*? Choose the most restrictive correct answer.

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n \log n)$
- E. $O(n^2)$

QUESTION 23

What does `intArray` look like after the method call `process(intArray)` where `intArray` begins as the array below?

1	2	3
4	5	6
7	8	9

A.

1	2	3
4	5	6
7	8	9

B.

0	1	2
3	4	5
6	7	8

C.

0	1	2
1	2	3
2	3	4

D.

1	2	3
2	3	4
3	4	5

E.

0	0	0
0	0	0
0	0	0

```
public static void process(int[][] array) {
    for (int i=0; i<array.length; ++i)
        array[i] = new int[] {i, i+1, i+2};
}
```

QUESTION 24

Suppose the integers from 1 to 8 were inserted into a binary search tree. What order of insertion would generate the tree shape to the right?

A. 1 2 3 4 5 6 7 8

B. 4 2 8 7 3 2 1 5

C. 4 8 5 3 2 1 6 7

D. 6 7 8 2 4 3 5 1

E. 6 7 2 8 1 3 4 5

QUESTION 25

What is the worst case running time of finding an item in a balanced binary tree with n elements? Choose the most restrictive correct answer.

A. $O(1)$ B. $O(\log n)$ C. $O(n)$ D. $O(n \log n)$ E. $O(n^2)$

QUESTION 26

What replaces **<*1>** in the code to the right to return numStars divided by maxStars and rounded to the nearest percent, that is, a number between 0 and 100?

- A. `100*Math.round(numStars/maxStars)`
- B. `100*Math.floor(numStars/maxStars)`
- C. `Math.floor(100*numStars/maxStars)`
- D. `Math.round(100*numStars/maxStars)`
- E. More than one of these

For the remaining questions, assume that **<*1>** has been filled in correctly.

QUESTION 27

Suppose there is a user-defined class `Movie` which represents movies. Which of these is the declaration of a data structure that can be used to associate movies with lists of their ratings? Given a movie, the data structure must provide efficient lookup of its ratings.

- A. `Map<List<Rating>,Movie> myRatings;`
- B. `Map<Movie,List<Rating>> myRatings;`
- C. `Movie<Map,Rating<List>> myRatings;`
- D. `List<Map<Movie<Rating>>> myRatings;`
- E. `List<Map<Movie,Rating>> myRatings;`

QUESTION 28

What is the output of the code below?

```
Rating r =
    new Rating("It was amazing!", 4.5, 5);
System.out.print(r.percentRating());
```

- A. It was amazing! B. 4.5
- C. 90 D. 4
- E. 0

QUESTION 29

What is returned by `process("12abCD")`?

- A. "12abCD" B. "12ABCD"
- C. "CD" D. " ABCD"
- E. The method does not compile

```
public class Rating {
    public Rating(String text,
                    double numStars, int maxStars) {
        if (numStars>maxStars || numStars<0 ||
            maxStars<1)
            throw new IllegalArgumentException();
        this.text = text;
        this.numStars = numStars;
        this.maxStars = maxStars;
    }

    public int percentRating() {
        return (int)(<*1>);
    }

    private String text;
    private double numStars;
    private int maxStars;
}
```

```
public static String process(String s) {
    StringBuffer sb = new StringBuffer();
    for (int i=0; i<s.length(); ++i)
        sb.append(Character.toUpperCase(
                                s.charAt(i)));
    return sb;
}
```

QUESTION 30

What sorting algorithm is implemented by the static method `Sorter.sort()`?

- A. Selection sort B. Insertion sort
- C. Quick sort D. Merge sort
- E. The method does not correctly sort

QUESTION 31

Which of the following best describes what can be passed as a parameter to `Sorter.sort()`?

- A. An array of items declared as `Object[]`
- B. An array of items that implement the `IntKey` interface declared as `Object[]`
- C. An array of items that implement the `IntKey` interface declared as `IntKey[]`
- D. An array of items that implement the `IntKey` interface declared as `Type[]` where `Type` is any class that implements `IntKey`
- E. Either C or D

QUESTION 32

Suppose that the array below is appropriate as a parameter to `Sorter.sort()`. Only the values returned by `key()` are shown rather than the full objects. What does the array look like after the third pass through the outer loop?

13	-4	10	5	6	18
----	----	----	---	---	----

- A.

13	13	13	5	6	18
----	----	----	---	---	----
- B.

-4	5	6	13	10	18
----	---	---	----	----	----
- C.

-4	5	6	10	13	18
----	---	---	----	----	----
- D.

-4	10	13	5	6	18
----	----	----	---	---	----
- E.

13	-4	10	5	6	18
----	----	----	---	---	----

```
public interface IntKey {
    public abstract int key();
}

public class Sorter {
    public static int find(IntKey[] a,
        int front, int back, int key) {
        int i = front + (back - front)/2;
        while (front < back) {
            if (a[i].key() < key)
                front = i+1;
            else
                back = i;
            i = front + (back - front)/2;
        }
        return front;
    }

    public static void sort(IntKey[] a) {
        for (int i=1; i<a.length; ++i) {
            int pos = find(a,0,i,a[i].key());
            IntKey data = a[i];
            for (int j=i; j>pos; --j)
                a[j] = a[j-1];
            a[pos] = data;
        }
    }
}
```

QUESTION 33

What is returned by `Double.parseDouble("31e2")`?

- A. 31.0 B. 3100.0 C. 310.0 D. 3.1 E. 0.31

QUESTION 34

Which of these static method calls returns 4?

- A. `fish("1fish2fishredfishbluefish")`
- B. `fish("1fish2fishredbluefishfish")`
- C. `fish("1fish2fishredfishbluefishes")`
- D. Both A and B
- E. Both B and C

```
public static int fish(String s) {
    String[] array = s.split("fish");
    return array.length;
}
```

QUESTION 35

What replaces **<*1>** in the code to the right to call the constructor for class A with parameter x?

- A. `super.x`
- B. `this.x`
- C. `super(x)`
- D. `this(x)`
- E. `A(x)`

```
public class A {
    public A(int x) {
        this.x = x;
    }
    public String toString() {
        return "" + x;
    }
    private int x;
}
```

QUESTION 36

What replaces **<*2>** in the code to the right to get the String returned by the `toString()` method from class A?

- A. `A.toString`
- B. `super.toString()`
- C. `super`
- D. `super.toString`
- E. `"" + x`

```
public class B extends A {
    public B(int x, int y) {
        <*1>;
        this.y = y;
    }
    public String toString() {
        return <*2> + y;
    }
    private int y;
}
```

QUESTION 37

What replaces **<*1>** in the code to the right to add all of the numbers in `intArray` and store the result in `sum`?

- A. `for (int i = 0; i<a.length; ++i) sum += i;`
- B. `for (int i : a) sum += i;`
- C. `for (int i : a) sum += a[i];`
- D. `sum += a;`
- E. Either A or B

```
int[] a = {1, 2, 3, 4, 5};
int sum = 0;

<*1>
```

QUESTION 38

What is the output of the code to the right?

- A. `%0(5.1f`
- B. `-2.5678`
- C. `(2.5678)`
- D. `(02.6)`
- E. An exception is thrown

```
double d = -2.5678;
System.out.printf("%0(6.1f", d);
```

QUESTION 39

What expression replaces **<*1>** in the code to the right, evaluating to true when j evenly divides i?

- A. (i % j != 0)
- B. (j / i != 0)
- C. (i % j == 0)
- D. (j % i == 0)
- E. (j / i == 0)

QUESTION 40

Assume **<*1>** is filled in correctly. What is output by the code below?

```
FactorGame f = new FactorGame(50);
try {
    f.makeMove(47);
    f.makeMove(39);
    f.makeMove(45);
    f.makeMove(32);
    f.makeMove(10);
}
catch(Exception e) {}
System.out.print(f);
```

- A. Player 0: 230
Player 1: 100
The game is not over
- B. Player 0: 148
Player 1: 101
The game is over
- C. Player 0: 116
Player 1: 81
The game is not over
- D. Player 0: 0
Player 1: 100
The game is over
- E. Player 0: 98
Player 1: 162
The game is not over

```
public class FactorGame {

    public FactorGame(int max) {
        values = new int[max+1];
        for (int i=0; i<max+1; ++i)
            values[i] = -1;
    }

    public void makeMove(int i) {
        if (i <= 0 || i >= values.length ||
            values[i] != -1 || done)
            throw new IllegalArgumentException();
        values[i] = turn;
        score[turn]+=i;
        turn = 1-turn;
        done = true;
        for (int j = 1; j<=i/2; ++j)
            if (<*1> && values[j]==-1) {
                values[j] = turn;
                done = false;
                score[turn] += j;
            }
    }

    public String toString() {
        return "Player 0: " + score[0] + "\n"
            + "Player 1: " + score[1] + "\n"
            + "The game is "
            + (done?"":"not ") + "over\n";
    }

    private int[] values;
    private int turn;
    private int[] score = new int[2];
    private boolean done = false;
}
```

Computer Science Answer Key

UIL Regional 2006

1. A	11. B	21. B	31. E
2. C	12. E	22. C	32. D
3. C	13. A	23. C	33. B
4. B	14. D	24. D	34. A
5. C	15. E	25. B	35. C
6. E	16. B	26. D	36. B
7. B	17. C	27. B	37. B
8. C	18. A	28. C	38. D
9. B	19. C	29. E	39. C
10. B	20. A	30. B	40. B